

KAIRÓS — Business Proposal

Accenture Innovation Challenge 2026 · Round 2 · BusinessIntelligence.ai

Kairós — the opportune moment. The distance between a number moving and someone acting on it is where the value sits, and it is measured in days.

1. Problem framing

Enterprises have solved metric *detection* and left metric *explanation* unautomated.

A dashboard can say North revenue fell 18.8%. It cannot say the fall is three enterprise accounts that stopped reordering after a warehouse missed its dispatch SLA ten days earlier — because that sentence lives in service tickets, call transcripts and a dispatch log that no BI tool reads. So the translation falls to an analyst, and it takes three to five days.

Three structural failures keep it there.

Noise blindness. Anomaly tools alert on deviation, not consequence. Flag everything and leaders mute you; flag on a fixed threshold and you miss the slow bleed that breaks the plan.

Correlation theatre. "Insight" features surface what moved alongside a metric. A leader cannot act on that — repricing, reallocating spend, re-staffing all assume a direction of cause. Fluent generated prose makes a weak correlation read as settled.

Organisational amnesia. The company solved this pattern fourteen months ago and knows which fix worked. That memory sits in a dead slide. Every recurrence is priced as new.

The measurable consequence is **decision latency**, not data latency. Pipelines refresh hourly; the explanation refreshes weekly.

Why this is not already solved

Roughly \$1.1B has been deployed across four adjacent categories — BI copilots, warehouse- native analysts, driver attribution and causal decision intelligence. Each solved a different third. Everyone automated the search. Nobody built the **standard of proof**, the **evidence outside the warehouse**, or the **memory of what worked**.

Empirically, the binding obstacle is elsewhere again. Olszak & Bartuś (*Procedia Computer Science* 270, 2025) interviewed twenty organisations across three sectors: **data availability and quality was the only barrier cited by every single respondent**, with source integration second at 18 of 20 — ahead of cost, talent and regulation. KAIRÓS treats that ranking as an instruction rather than a footnote.

2. Solution design

An explanation layer that sits on top of the BI stack the client already owns. It does not replace Power BI, Tableau or the warehouse; it attaches to the existing semantic layer and returns narratives into the tools people already open.

The engine

Ten stages, each defined as much by what it refuses to do as by what it computes: SEMANTIC → FITNESS → DRIFT → SIFT → SPLIT → SOURCE → PROPAGATE → FORECAST → SOLVE → NARRATE.

The four governors

The Evidence Ladder — Bradford Hill's viewpoints (1965) operationalised on Pearl's causal hierarchy (CACM 2019). L0 co-movement · L1 precedence and dose–response · L2 independent corroboration · L3 counterfactual. **Nothing below L2 is ever phrased as a cause.** Conflict is measured: evidence supporting a rival theme above a 0.4 ratio marks a hypothesis *contested* and blocks L2.

The Ambiguity Protocol — when explanations tie, the engine returns *Confirmed / Competing / Unknown*, shows the rivals side by side, and **prices the cheapest test that separates them**. Showing rivals by default is a cognitive forcing function, shown to reduce over-reliance on incorrect AI suggestions (Bućinca et al., CSCW 2021).

The Numeric Guard — every figure in generated text is parsed and matched against the closed evidence packet. Unverifiable numbers fail the narrative.

The Decision-Rights Router — each recommendation is routed to a human–AI collaboration mode derived from evidence × value at risk against the owner's authority limit × reversibility, following the human–AI decision-division taxonomy in Prasanth et al. (*IJACSA* 14(6), 2023). Full delegation is reserved and never assigned.

Division of labour with the model

The design follows the evidence: models are strong at knowledge-based causal *proposal* (Kıcıman et al., TMLR 2024) and near-random at inferring causation *from correlation* (Corr2Cause, ICLR 2024). So the model proposes and phrases; deterministic logic, SQL, statistics, a learned ranker and causal inference decide and compute.

Enforced, not asserted: **91–100% of reasoning steps are non-LLM**, and the engine produces a complete answer with the model switched off entirely.

3. Target users

PERSONA	WHAT CHANGES	DELIVERY
CFO / Finance	Walks into review with cause, size, owner and a costed next step rather than four correlations	Monday exec brief
Regional Sales Manager	Named accounts and a specific action they own, not a regional average	CRM task + digest
Supply Chain Lead	The warehouse root cause in operational units — no commercial data	Ops standup board
Data Analyst	Stops being a lookup service; becomes curator of the causal graph and the playbooks	Analyst workbench

The analyst is both the biggest winner and the quality control: they grade every diagnosis, and that grade gates the rollout.

4. Business case and impact

Value comes from collapsing the distance between a number moving and someone acting.

LEVER	MECHANISM	BASIS IN THE PROTOTYPE
Recovered leakage	Acting in week 1 rather than week 2 on service-driven churn	S1: ₹2.15 Cr movement, 87.6% attributable, playbook recovery rate 71% measured
Analyst capacity	"Why did X move" fire-drills automated	Each incident is 2–4 analyst days today
Avoided wrong actions	Abstention prevents intervening on the wrong cause	S2: two rivals, both L1 — a confident engine would have picked one
Compounding memory	Effect sizes re-estimated from realised outcomes	Playbook accuracy improves with each closed loop

Cost. ≈450 ms end-to-end; ≈80 ms and zero tokens on a cache hit. One model call at ~2.5k in / ~200 out ≈ **₹0.30 per insight**. Retrieval is scoped to a single segment and window, so evidence gathering stays affordable per insight rather than per corpus.

Scale. At 4.5M order rows — a hundred times the demo estate — the three queries the engine actually runs complete in 1.9–3.2 ms.

Success metrics — deliberately not insight volume, which is the metric that manufactures alert fatigue:

1. **Diagnosis precision** — analysts blind-grade narratives; this gates rollout.
2. **Decision latency** — days from movement to owned action.
3. **Recommendation calibration** — predicted against realised recovery.

Defensibility. The causal graph and the playbook library are built from the client's own history. A competitor's generic model cannot copy them, and they strengthen with every incident. Packaged per sector they become a reusable delivery accelerator rather than a one-off build.

5. Phased roadmap

PHASE	SCOPE	EXIT CRITERION
1 — Prove precision (0–3 mo)	One P&L metric, one function. SEMANTIC + FITNESS + SIFT + SPLIT on the existing semantic layer. Narratives reviewed, never auto-published.	Analyst-graded precision clears an agreed bar. Precision is the only gate.
2 — Add evidence and memory (3–6 mo)	Connect unstructured sources with entitlements and redaction. Seed playbooks from past incidents. Enable the Evidence Ladder and abstention.	≥60% of material movements reach L2+; abstention rate stable and defensible
3 — Close the loop (6–12 mo)	Multi-KPI, multi-persona, outcome logging, the learned ranker trained on the client's resolved episodes. Recommendations remain proposed .	Recommendation calibration measurable; decision latency down
4 — Productise (12 mo+)	Industry-tuned graph and playbook packs as a delivery accelerator	Reuse across engagements

6. Key risks and mitigations

RISK	MITIGATION — BUILT IN, NOT BOLTED ON
Confident but wrong	L2 evidence floor; explicit <i>Unknown</i> ; mandatory citation; numeric guard rejects unverifiable figures
Spurious causes	Curated causal graph blocks inadmissible mechanisms; placebo must validate before L3; automated search treated as exploratory, not confirmatory
Poor data quality	A fitness gate runs <i>first</i> across five dimensions and may halt the run. Partition- and class-level completeness catches a load that drops one kind of row while totals hold
The ground moves	Drift monitored separately for data and model; a ranker scoring outside its training support has its authority withdrawn for that run
Definitional disputes	Competing KPI definitions reconciled once, before analysis, with the resolution and the gap recorded
Over-trust	Rival hypotheses shown by default; actions stay <i>proposed</i> ; full machine delegation reserved and never assigned
Sensitive evidence	Row, column and domain entitlements applied to data <i>before</i> prompt assembly; PII redacted pre-retrieval; withheld items surfaced as a count
Regulatory	EU AI Act Art. 12 post-hoc reconstruction satisfied by full lineage; Art. 14 human oversight by never auto-executing. India's DPDP obligations (May 2027) make purpose limitation on call data a design constraint
Financial reporting exposure	A narrative reaching a board pack sits inside ICFR scope, so SPLIT is deterministic arithmetic and narratives ship marked decision-support until a human validates
Platform commoditisation	We deliberately do not build text-to-SQL or a semantic layer — now free platform features. The moat is the standard of proof, the evidence outside the warehouse, and the memory

7. What the prototype demonstrates

Against a synthetic estate with deliberately planted ground truth, KAIROS:

- **recovered** the buried service failure at L3 — DiD -20.6%, $p < 0.0001$, placebo clean — and independently rediscovered the 10-day mechanism lag from the causal graph;
- **named** the three affected accounts and separated the 87.6% volume driver from the mix driver with an identity that closes to under ₹1;
- **abstained** where two explanations were confounded by design, and priced a ₹1.8L, 14-day separating test instead of guessing;
- **refused** to model a 19-day-old series, falling back to a peer prior;
- **blamed the pipeline**, not the business, when late-shipment rows stopped loading;
- **reconciled** two competing definitions of net revenue and reported the gap;
- **denied** a regional manager a query outside their entitlement before any SQL ran, and produced a rupee-free narrative for a persona denied financial columns;
- **withdrew** its own learned ranker's authority when inputs fell outside its training support;
- **caught its own hallucination** — an injected ₹4.2 Cr figure failed the numeric guard and the deterministic narrative was published instead.

125 automated tests assert these outcomes against the ground-truth file.